

EE100 – Assignment 2

Question – 1:

- 12.11** For the three-phase circuit of Fig. 12.44, $\mathbf{I}_{bB} = 30 \angle 60^\circ$ A and $\mathbf{V}_{BC} = 220 \angle 10^\circ$ V. Find $\mathbf{V}_{an}$, $\mathbf{V}_{AB}$, $\mathbf{I}_{AC}$, and $\mathbf{Z}$.

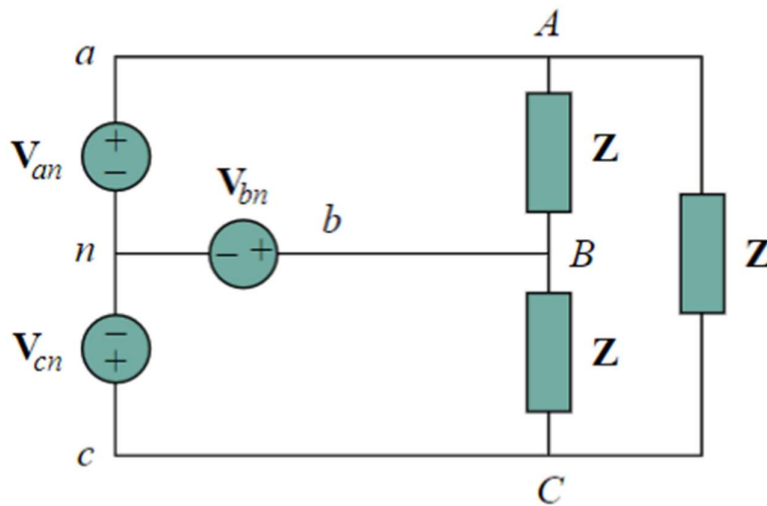


Figure 12.44 For Prob. 12.11.

Question – 2:

- 12.23** In a balanced three-phase Δ -Y circuit, the source is connected in the positive sequence, with $\mathbf{V}_{ab} = 220 \angle 20^\circ$ V and $\mathbf{Z}_Y = 20 + j15 \, \Omega$. Find the line currents.

Question – 3:

- 12.25** A balanced wye-connected load absorbs a total power of 5 kW at a leading power factor of 0.6 when connected to a line voltage of 240 V. Find the impedance of each phase and the total complex power of the load.

Question – 4:

- 12.29** The total power measured in a three-phase system feeding a balanced wye-connected load is 12 kW at a power factor of 0.6 leading. If the line voltage is 208 V, calculate the line current I_L and the load impedance Z_Y .

Question – 5:

- 12.39** The following three parallel-connected three-phase loads are fed by a balanced three-phase source.
- Load 1: 250 kVA, 0.8 pf lagging
 - Load 2: 300 kVA, 0.95 pf leading
 - Load 3: 450 kVA, unity pf
- If the line voltage is 13.8 kV, calculate the line current and the power factor of the source. Assume that the line impedance is zero.

Solutions:

Q1: $127 \angle 100^\circ$ V, $220 \angle 130^\circ$ V, $17.32 \angle 150^\circ$ A, $12.7 \angle -80^\circ \Omega$

Q2: $5.081 \angle -46.87^\circ$, $5.081 \angle -166.87^\circ$, $5.081 \angle 73.13^\circ$ A

Q3: $4.15 - j 5.53 \Omega$, $5000 - j 6667$ VA

Q4: 55.51 A, $1.298 - j 1.731 \Omega$

Q5: 40.42 A (rms), 0.9677 (lagging)